MES COLLEGE OF ENGINEERING-KUTTIPPURAM
DEPARTMENT OF COMPUTER APPLICATIONS
20MCA246 – MAIN PROJECT

PRO FORMA FOR THE APPROVAL OF THE FINAL SEMESTER PROJECT

(Note: All entries of the pro forma of approval should be filled up with appropriate and complete information. Incomplete Pro forma of approval in any respect will be rejected.)

Project Proposal Number : 01
(Filled by the Department)

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Academic Year : 2024-2026
Year of Admission : 2024
Admission Number : 18921
Roll Number : 16
Register Number : MES24MCA-2016

1. Name of the Student (in BLOCK LETTERS) : FATHIMATH SANA C V
2. Name of the Organization : __________
3. Address of the Organization : __________
   Telephone No. : __________ Company E-Mail : __________
4. Name of the External Guide : __________
   Mobile No. : __________ E-Mail : __________
5. Title of the Project : ELDERLY MONITORING SYSTEM
6. Name of the Guide (Internal-Departmet) : PRIYA J D

Date : 05/01/2026 Signature of the Student:

Comments of The Project Guide

Initial Submission :

Approval Status : Approved / Not Approved Dated Signature of Guide HOD

First Review :

Second Review :

Third Review :

Comments of The Project Coordinator

Initial Submission:

First Review : Second Review: Third Review:

Dated Signature of Project Coordinator:

Elderly Monitoring System

Abstract

The Elderly Monitoring System is an advanced healthcare and safety management solution designed to address the growing challenges associated with aging populations. Elderly individuals are highly vulnerable to falls, health complications, and security threats, especially when living alone or in assisted environments. Falls are one of the leading causes of serious injuries and accidental deaths among senior citizens, making continuous monitoring and immediate response essential. This system leverages Machine Learning techniques, specifically Support Vector Machine (SVM) and Decision Tree algorithms, to accurately analyze human activities and distinguish between fall and non-fall events.

The system continuously monitors elderly movements and behaviors through sensor or camera-based inputs and processes the data using trained machine learning models. When a fall is detected, the system immediately generates alerts and notifications that are sent to caretakers, family members, or medical personnel, along with relevant details to ensure prompt emergency response. This automated alert mechanism significantly reduces response time and minimizes the risk of severe health outcomes.

Beyond fall detection, the Elderly Monitoring System offers a comprehensive healthcare management platform. It includes pill reminders to ensure medication adherence, doctor appointment scheduling with notifications, prescription management, chatbot assistance for health-related queries, and personalized health tips. These features help elderly users maintain regular healthcare routines and reduce dependency on manual supervision.

To further enhance safety, the system integrates a visitor logbook with face recognition technology. This module identifies visitors as known or unknown individuals and automatically records entry details. Caretakers can remotely access visitor logs, improving security, preventing unauthorized access, and providing peace of mind to both elderly users and their families.

The system is developed using modern web and mobile technologies, with a robust backend powered by Python-based frameworks and a MySQL database. The modular design ensures scalability, reliability, and ease of use across multiple platforms. By combining real-time monitoring, intelligent decision-making, healthcare assistance, and security features, the Elderly Monitoring System provides a holistic solution that enhances safety, independence, and quality of life for elderly individuals. This project demonstrates the effective application of machine learning and intelligent systems in addressing real-world healthcare and social challenges.